

Does Economic Concept Selection Matter for Monetary Policy Shock Identification?

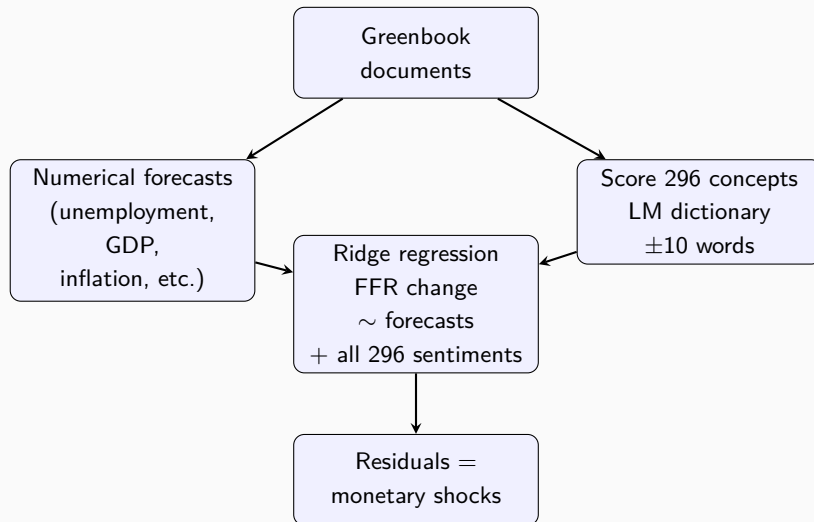
Evidence from Aruoba & Drechsel (2024)

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Monetary Policy & Financial Markets

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AD(2024) Scores Sentiment with a Dictionary



First: We Replicated the Original Results

Table 2 — Forecast Error Predictability

Horizon	Our R^2	Paper
Current Q	0.045	0.045
1Q ahead	0.149	0.149
1Y ahead	0.248	0.248
2Y ahead	0.208	0.208

PC1 of 296

sentiments $\rightarrow$ forecast errors.

All horizons match exactly.

Table 3 — First-Stage Fit

Specification	Vars	Our R^2	Paper
(4) Fc + sentiments	429	0.64	0.65
(7) Full spec	3,258	0.96	0.94

Paper:

R glmnet ($\alpha = 0$). Ours: sklearn RidgeCV.
 ~ 0.01 diff = CV lambda grid variation.

Both tables reproduced. Data and pipeline verified.

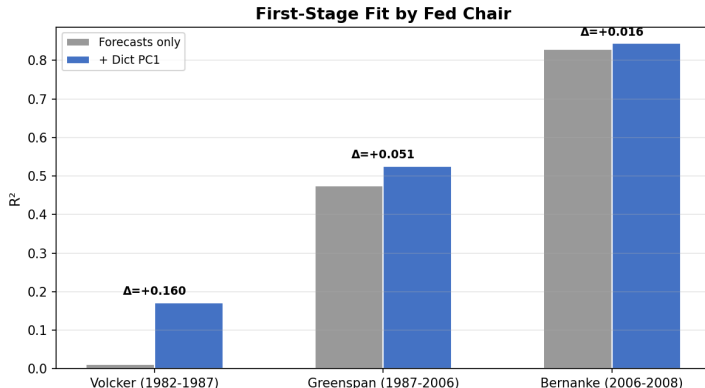
We Reproduce Table 3 Across All Specifications

Specification	Vars	Our R^2	Paper
(1) Romer-Romer OLS (basic forecasts)	17	0.46	0.50
(2) Ridge with extended forecasts	132	0.50	0.55
(3) + nonlinear forecasts	298	0.51	0.51
(4) + all 296 sentiments (linear)	445	0.64	0.65
(5) + nonlinear sentiments	890	0.74	0.75
(6) + 4 lags of sentiments (linear)	1,629	0.69	0.87
(7) Full spec: + lags, nonlinear	3,258	0.96	0.94

Ridge, 10-fold CV (R glmnet $\alpha = 0$ vs. sklearn RidgeCV). All specs reproduced within CV margin.

Specs (4)–(7) include the 296 sentiment indicators — these are what we test below.

Sentiment Adds Value Under Volcker, Greenspan, and Bernanke



FFR change $\sim$ extended forecasts (gray) vs. FFR change $\sim$ extended forecasts + Dict PC1 (blue)

Volcker (1982–87)
 $n = 39$. Discretionary. $\Delta R^2 = +0.16$

Greenspan (1987–06)
 $n = 143$. Great Moderation. $\Delta R^2 = +0.05$

Bernanke (2006–08)
 $n = 22$. Pre-crisis. $\Delta R^2 = +0.02$

The 296 Concepts Were Hand-Picked by the Authors

How the list was built

1. Pull n-grams from 787 Greenbook PDFs
2. Sort by document frequency
3. **Authors manually pick** which ones are “economic concepts”
4. Final list: 296

A few of the 296 ...

advanced foreign economies
business confidence
consumer sentiment
domestic final purchases
economic expansion
emerging market economies
equity price
foreign net purchases

*If someone else did step 3,
would they pick the same 296?*

We Replaced the Human Curator with an LLM

Same 2,000 candidate n-grams from the same 787 texts.

LLM reads each one: *“economic concept or not? Which category?”*

The LLM picked 595 concepts. (The original authors picked 296.)

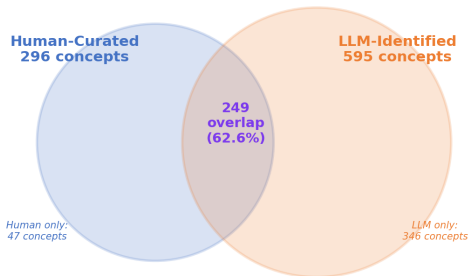
By category

Real activity	204
Financial	150
Prices	99
International	51
Labor	48
Fiscal/other	43
Total	595

- Nearly **twice as many**
- More coverage of financial, price, and international terms
- Did it agree with the human list on specifics?

Human and LLM Agreed on Only 62.6% of Concepts

Who Chooses the Concepts?



Human: 296

LLM: 595

Overlap: 249

Same texts, same 2,000 candidates. Different classifier. **37.4% disagreement.**

AD(2024): "n-gram frequency analysis + manual screening."

The frequency step is mechanical. The screening is not.

Humans Chose Specific Terms; the LLM Chose Broad Ones

Only human picked

specific, concrete

domestic final purchases
district banks
business confidence
consumer sentiment
drilling activity
employment cost

Both picked (249)

agreed by both

inflation
unemployment
gdp growth
housing starts
exports
credit

Only LLM picked

broad, general

prices
markets
labor
federal reserve
growth
energy

Human: specific. LLM: broad. **Different taste, not right vs. wrong.**

Prompt: "You are an economist. For each term, decide: is this an economic concept?" Yes/No + category. 50 terms per batch.

Temperature = 0.1.

If Concept Selection Is Subjective, How Much Does It Matter?

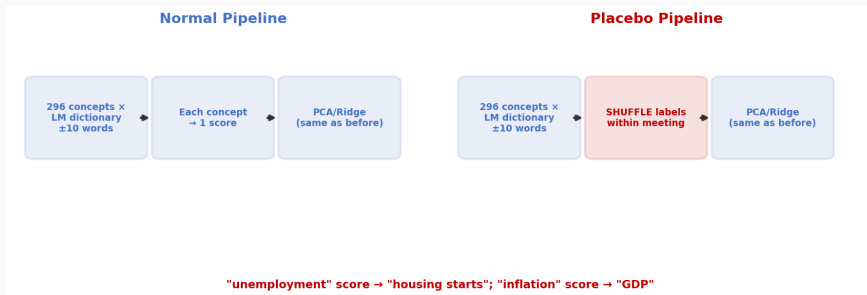
Concept selection is fundamentally **subjective**.

Yet the headline — $R^2 = 0.94$ — rests on
the exact list those particular authors settled on.

**How much of the result is about the words chosen,
and how much is about the scoring method itself?**

An extreme test: **shuffle all concept labels. Re-run everything.**

We Shuffle All Concept Labels, Then Re-Run the Regression



For each meeting, randomly permute the 296 scores across columns.

“Unemployment” gets “housing starts”’ score. “Inflation” gets “GDP”’s score.

Same scoring. Same regression. Same number of variables. **Only the labels change.**

The Shuffle Scrambles Scores Within Each Meeting

Before shuffle (correct labels)

	unemp.	inflat.	GDP	credit
Meeting 1	+0.3	−0.1	+0.5	−0.2
Meeting 2	−0.4	+0.2	−0.1	+0.0
Meeting 3	+0.1	−0.3	+0.4	−0.1

After shuffle (labels permuted)

	unemp.	inflat.	GDP	credit
Meeting 1	−0.1	−0.2	+0.3	+0.5
Meeting 2	+0.0	+0.2	−0.1	−0.4
Meeting 3	−0.3	−0.1	+0.1	+0.4

Within each row, the **same 4 numbers** — just in different columns.

```
for i in range(n):    X_shuffled[i, :] = rng.permutation(X_sent[i, :])
```

With 429 Variables, Shuffling Labels Changes Nothing

Specification

132 extended forecasts + 296 sentiment indicators $\rightarrow$ Ridge, 10-fold CV. 429 vars.

Correct labels

$$R^2 = 0.64$$

Shuffled labels (mean of 100 permutations)

$$R^2 = 0.65$$

No drop.

All 296 Scores Point the Same Way Within a Meeting

Inside one FOMC meeting...

If staff is **OPTIMISTIC**:

"employment continues to improve"
"inflation remains contained"
"consumer spending is robust"
"investment picks up"

If staff is **PESSIMISTIC**:

"unemployment rises further"
"inflationary pressures build"
"spending weakens"
"investment remains sluggish"

**All 296 concepts point in the SAME direction.
PC1 = the common direction. Labels do not matter.
296 concepts = 296 sampling points for one document-level tone.**

With 3,224 Variables, Shuffling Drops R^2 by Only 0.03

Specification

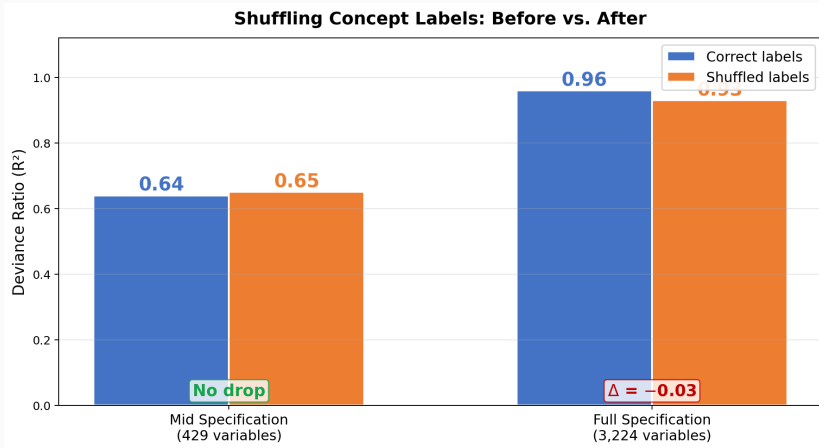
```
X = np.column_stack([X_forecasts, X_forecasts**2,  
                    X_sentiments, X_sentiments**2,  
                    X_sentiments_lag1, ..., X_sentiments_lag4,  
                    X_sentiments_lag1**2, ..., X_sentiments_lag4**2])  
ridge = RidgeCV(cv=10).fit(X, y)
```

Correct labels: $R^2 = 0.96$

Shuffled labels (mean of 50): $R^2 = 0.93$

Dropped by 0.03.

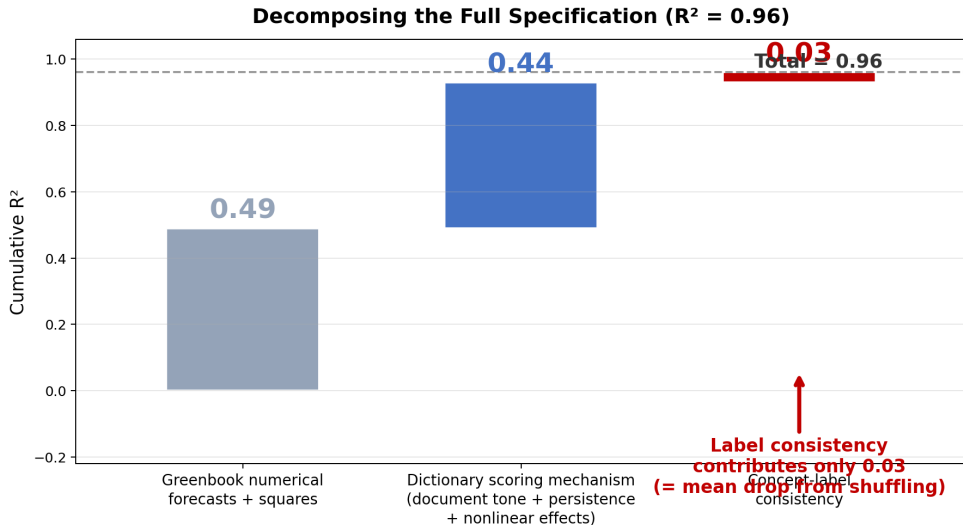
Shuffling Labels: The Full Picture



Mid spec: no effect. Full spec: only 0.03 lost.

The concept labels themselves contribute almost nothing to the fit.

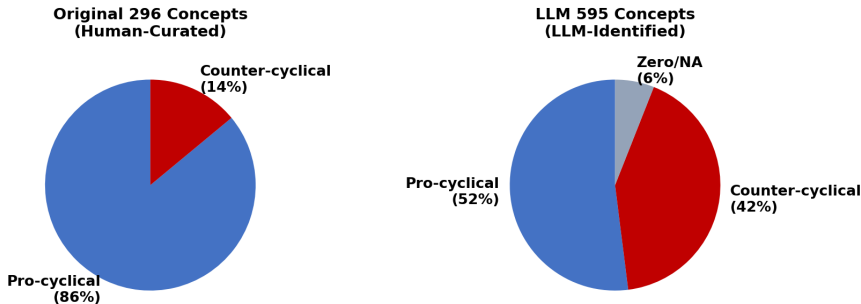
The Scoring Method Contributes 10× More Than Label Identity



Of the 0.96, only 0.04 depends on which word is which.

The Original 296 Concepts Are Mostly Pro-Cyclical

Why the Placebo Test Passed: Most Original Concepts Point the Same Way



86% of the original 296 move with the cycle. Only 14% go the other way.

The LLM's 595 are much more balanced (52% vs. 42%).

The placebo test passed partly because the original list is already one-sided.

What Does “Pro-Cyclical” Mean — and Why Does It Matter?

Definition

A concept is **pro-cyclical** if its sentiment score is positively correlated with PC1 (the common factor). When the economy is good, its score is positive. When bad, its score is negative. **It moves with the cycle.**

Example: pro-cyclical concepts

Concept	Corr. with PC1	
business activity	+0.73	Economy
firms	+0.72	
manufacturing	+0.71	
employment	+0.69	
construction	+0.68	

improves → positive language around these concepts
→ positive score.

Example: counter-cyclical concepts

Concept	Corr. with PC1	
output gap	−0.33	Economy
potential output	−0.32	
uncertainty	−0.25	
federal debt	−0.17	
fiscal policy	−0.15	

improves → cautious or negative language around these concepts.

How we determine this

For each concept, correlate its 204-meeting sentiment time series with PC1.

Positive correlation = pro-cyclical. Negative correlation = counter-cyclical.

Devesh Thavalam: “Apply the Shuffle Test to the LLM List”

The argument

The human list is 86% pro-cyclical — shuffling labels within a meeting barely changes the score distribution.

The LLM list is 52%/42% pro-/counter-cyclical — more balanced, so shuffling should be **more disruptive IF labels matter**.

Does the shuffle test produce the same answer with the LLM list?

We replicate the identical procedure on the LLM-generated 595 concepts.

What we test

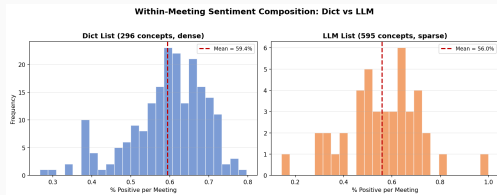
(1) Shuffle LLM sentiment columns within each meeting. (2) Recompute PC1. (3) Re-run Ridge (Forecasts + PC1).

100 independent permutations. Compared against the same test on the human 296.

The LLM List Has More Within-Meeting Variation — But Is 99% Sparse

Composition comparison

	Human	LLM
Concepts	296	595
Non-zeros/meeting	296.0	5.3
% Positive (mean)	59.4%	56.0%
% Positive (std)	0.099	0.155
% Positive (min)	26.7%	14.3%
% Positive (max)	79.7%	100.0%



LLM list (right): higher variance, wider range, long left tail.

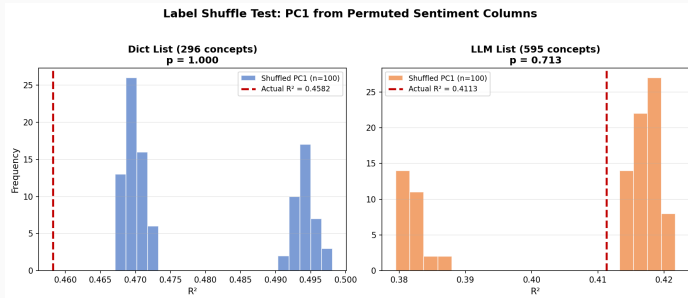
Critical caveat

The LLM list is **99% sparse** — only ~5 concepts receive non-zero scores per meeting.

This means the shuffle test only permutes ~5 labels per meeting, not 595.

The sparsity is by design (LLM only scores concepts the document actually discusses), but it limits permutation-based inference.

PC1 Shuffle Test: Labels Don't Matter for Either List



Human (296)

Actual $R^2 = 0.458$

Shuffle mean = 0.479 (**higher!**)

$p = 1.000$

LLM (595)

Actual $R^2 = 0.411$

Shuffle mean = 0.407 (-0.004)

$p = 0.713$

Neither list shows label-dependence. The human list actually fits *better* with shuffled labels.

LLM PC1 is weaker overall ($\Delta R^2 = +0.032$ vs. $+0.079$ for dict) due to extreme sparsity.

Full-Column Shuffle: Using All Sentiment Columns Directly

Procedure

Skip PC1 reduction. Shuffle columns within each meeting. Re-run Ridge directly on

```
np.column_stack([X_forecasts, X_shuffled_sentiments])
```

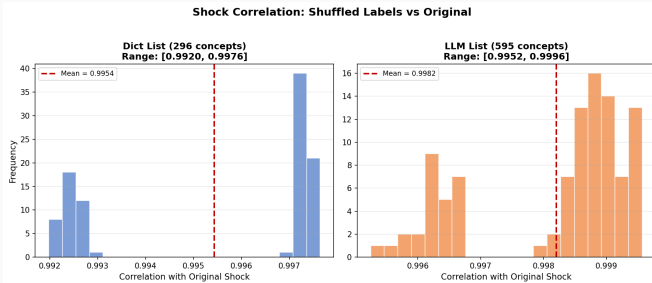
100 permutations. No lags, no squares — pure concept-level signal.

	Actual R^2	Shuffle Mean	Std	p
Human (296 cols)	0.656	0.692	0.045	0.624
LLM (595 cols)	0.609	0.626	0.0003	1.000

Human: modest increase in R^2 , meaningful variation across shuffles.

LLM: near-zero variation — shuffling 5 non-zero scores across 595 columns changes almost nothing.

Shock Correlation: Both Lists Invariant to Label Permutation



	Mean r	Std	Range
Human (296)	0.9954	0.0024	[0.9920, 0.9976]
LLM (595)	0.9982	0.0012	[0.9952, 0.9996]

Both lists produce shocks essentially identical to the original after shuffling.

The LLM list is *even more* invariant because extreme sparsity means shuffling barely alters the data matrix.

The theoretical ideal (balanced, dense concept list) is not what the LLM produced — making this a conservative test.

Interpretation: Coverage, Not Labels

Devesh's premise was correct in principle

The LLM list has more balanced composition → shuffle test should be more informative.

The data confirms: LLM list has higher within-meeting score variance (std 0.156 vs. 0.099), wider range.

But the LLM list is also 99% sparse — a design feature, not a bug

The LLM only scores concepts the document actually discusses. Zeros mean “not mentioned.”

This sparsity means only ~ 5 labels are permuted per meeting — the shuffle test is underpowered.

Yet the answer is unambiguous

Neither list shows label-dependence. Shock correlations > 0.99 for both.

The human list's shuffle R^2 actually *increases* — the original labels may add noise, not signal.

Labels are incidental. The number of sampling points is what matters.

Concept Selection Matters — At the Level of List Composition

The method is robust across regimes

Sentiment adds incremental R^2 under Volcker (+0.16), Greenspan (+0.05), and Bernanke (+0.02).

The AD(2024) pipeline works. The question is **why**, and **where its limits are**.

The concept list is subjective — but that barely matters for the result

Human vs. LLM curation: only 62.6% agreement. Yet shuffling all 296 labels drops R^2 by only 0.03.

Why? Because 86% of the original concepts are pro-cyclical

Most point the same way within any meeting. The common signal dominates regardless of labels.

New: LLM list shuffle test (Devesh Thavalam Rec. #3)

Replicated the shuffle test on a 595-concept LLM-generated list with more balanced composition.

Same result: shock correlations > 0.99 . Neither list shows label-dependence.

LLM list is 99% sparse — a conservative test, but the conclusion holds. **Coverage, not labels.**

Thank You